Scalability & Future Plan

Date	**04 July 2026**
Team ID	**xxxxxxxxxx**
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the OptiCrop solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the roadmap for enhancing and evolving the project beyond its current state. The system is designed to support increasing users and larger agricultural datasets without affecting prediction accuracy.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address
1	Local deployment only — not cloud hosted	App is only accessible on local machine — cannot be shared publicly	High
2	Limited to single ML model (Logistic Regression)	May not perform optimally for all crop types and regional conditions	Medium
3	No real-time weather API integration	Farmers must manually input climate data instead of automatic fetching	High

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single user — local Flask dev server	Deploy on cloud platform (AWS/Heroku/Render) with load balancing for multiple concurrent users
2	Data Storage	CSV file with 2200 records locally	Migrate to PostgreSQL/MongoDB database for larger datasets, real-time updates, and user history storage

3	Performance	Single ML model — Logistic Regression	Implement ensemble models (Random Forest, XGBoost) and AutoML for improved prediction accuracy
4	Security	No authentication — open access	Add user authentication (JWT/OAuth), input sanitization, and HTTPS for secure production deployment

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Cloud Deployment on Heroku/Render + Real-time Weather API integration for automated climate data	3–6 months	Public accessibility for farmers across India; eliminates manual climate input
Phase 3	Mobile App (Android/iOS) + Regional Language Support (Telugu, Hindi) + Advanced ML models (Random Forest, XGBoost)	6–12 months	Wider farmer adoption; improved prediction accuracy across diverse agricultural regions
Phase 4	IoT Sensor Integration for automated soil parameter detection + AI-powered crop disease detection using image classification	12–18 months	Fully automated smart farming system; real-time crop health monitoring and disease prevention

Expected Impact

- Better agricultural productivity
- Increased farmer income
- Efficient resource utilization
- Sustainable farming